

1. TITLE

Implement Modules in NodeJS

2. AIM

To implement and use custom modules in NodeJS for performing reusable mathematical operations.

3. OBJECTIVES

1. To understand the NodeJS module system.
2. To create a custom JavaScript module.
3. To export functions from a module.
4. To import a custom module using require().
5. To execute reusable functions from another file.

4. THEORY

NodeJS supports modular programming in which related functionality can be placed in separate files. A module can export functions or values and another file can import them. This improves code organization, reuse and maintainability.

5. ALGORITHM/PROCEDURE

1. Create a file named math.js for mathematical functions.
2. Define add(), multiply() and square() functions.
3. Export the functions using module.exports.
4. Create app.js and import the module using require('./math').
5. Call the exported functions with suitable values.
6. Display the returned values in the NodeJS terminal.
7. Run the program using node app.js and verify the output.

6. PROGRAM

```
// math.js
function add(a, b){
    return a + b;
}

function multiply(a, b){
    return a * b;
}

function square(a){
    return a * a;
}

module.exports = {
    add,
    multiply,
    square
};

// app.js
const math = require("./math");

const a = 8;
const b = 5;

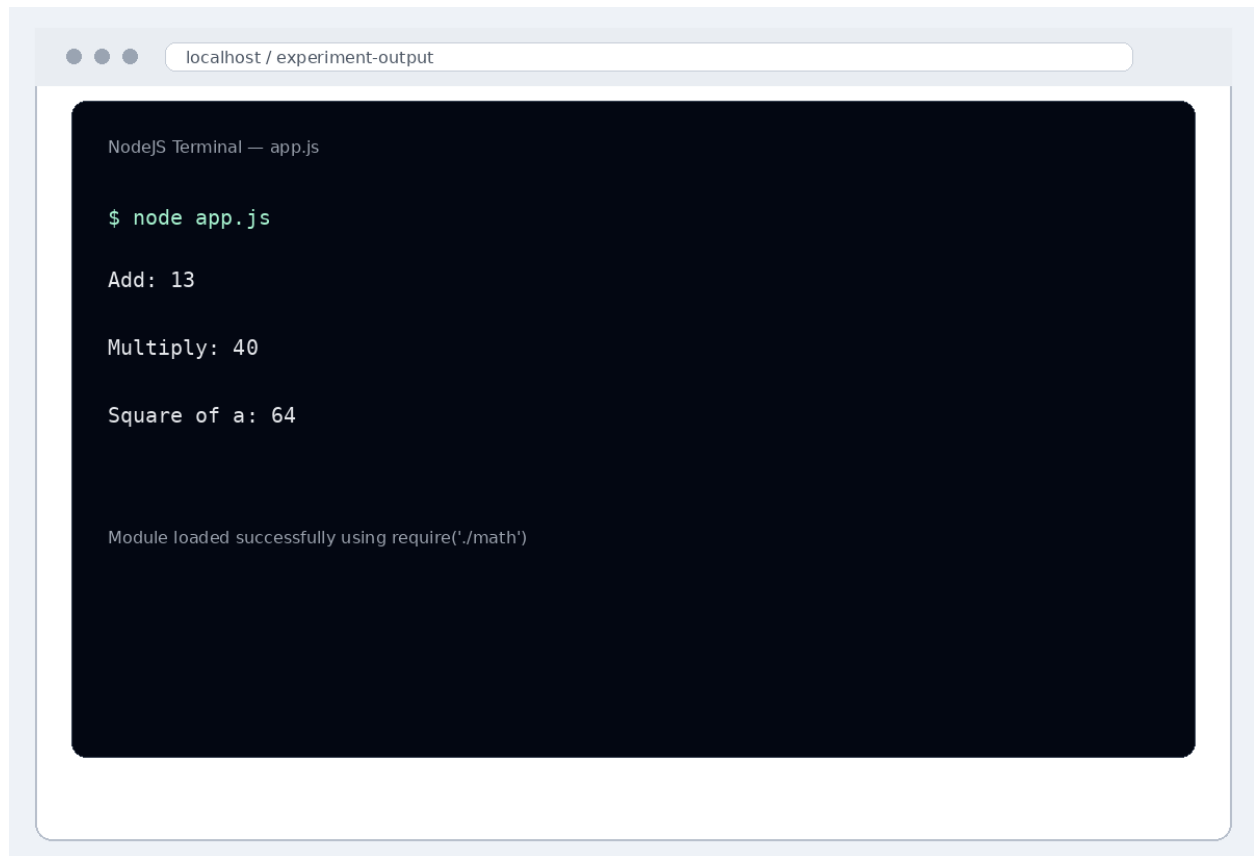
console.log("Add:", math.add(a, b));
console.log("Multiply:", math.multiply(a, b));
console.log("Square of a:", math.square(a));
```

7. CODE EXPLANATION

Component	Purpose
function add()	Returns the sum of two numbers.
function multiply()	Returns the product of two numbers.
function square()	Returns the square of a number.
module.exports	Exports functions so another file can use them.
require('./math')	Imports the custom math module into app.js.
console.log()	Displays the calculated values in the terminal.

8. OUTPUT

Browser Output (Screenshot):



9. RESULT

Thus, a custom NodeJS module was successfully created and imported. The exported mathematical functions were reused from app.js and the expected results were displayed in the terminal.